UWAIS MANSURI

AI Engineer | GenAI Platforms · AWS · Python

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SUMMARY

AI engineer with 4+ years of cloud and backend engineering experience building and operating production AWS healthcare systems. Currently developing a GenAI service used daily by ~3,000 pharmacists and nurses, with hands-on ownership across Python orchestration, LLM security and evaluation, Terraform, reliability, observability, and PHI-safe integration.

EXPERIENCE

Tata Consultancy Services (TCS) Feb 2022 - Present

GenAI Platform Engineer | Cigna Healthcare (Evernorth) Apr 2026 - Present

- Architected and productionized an AWS serverless GenAI service that gives ~3,000 daily clinical users role-specific pre-encounter summaries from patient data via parallel retrieval, a de-identified LLM boundary, and deterministic post-processing.
- Reduced generated output by about 700 tokens per request in a measured workflow through deterministic Python extraction; enabled prompt caching for reusable instructions and context to reduce repeated input processing and inference cost.
- Designed a 49-case OWASP LLM Top 10-aligned adversarial and API negative-testing suite covering prompt injection, data leakage, unsafe output, resource exhaustion, schema validation, and JWT authentication; fixed a malformed-token crash and passed all defined scenarios using synthetic non-PHI data and live model calls.
- Engineered deadline-aware execution with four parallel data fetches, concurrent LLM calls, timeouts, circuit breakers, and tiered fallbacks; corrected parallel tool-call output loss through exhaustive PydanticAI execution and deterministic merge and deduplication.
- Codified GenAI infrastructure in Terraform across Lambda, API Gateway, IAM, KMS, Secrets Manager, networking, DynamoDB, CloudWatch alarms, and scheduled Provisioned Concurrency; delivered reusable CI/CD with structured logging and OpenTelemetry/LangSmith tracing.

Cloud and Backend Platform Engineer | Cigna Healthcare Jan 2023 - Mar 2026

- Owned development and production support for an event-driven AWS platform shared by four enterprise applications, using Python, Lambda, API Gateway, S3, SQS/DLQs, DynamoDB, Aurora PostgreSQL, OpenSearch, and Amazon Comprehend.
- Automated email intake from Pega by processing .eml files and attachments through S3, SQS, and Lambda; invoked a custom Comprehend model to classify RFP and non-RFP submissions, deduplicate events, and route downstream workflows.
- Operated CDC and transformation pipelines that normalized JSON events and replicated data from raw to integrated tables; diagnosed Lambda failures and database check-constraint violations that interrupted replication.
- Sustained 99.9% platform availability and met 98% of operational SLAs by tuning Lambda concurrency and throttling, resolving SQS delays and DLQs, and monitoring CloudWatch, Grafana, Splunk, and Dynatrace synthetics.
- Led EKS release validation and pod recovery during production deployments; coordinated incident RCA and implemented VPC, subnet, and load-balancer improvements to strengthen release stability and recovery.
- Secured API access with a scheduled Okta token-refresh workflow using EventBridge, Lambda, Secrets Manager, and DynamoDB, plus group-based authorization through a Lambda Authorizer.

Data Engineer | TCS Feb 2022 - Dec 2022

- Designed dimensional data models (fact and dimension tables) and built Python and SQL ETL jobs with data-quality checks for an organization-wide data warehouse feeding business reporting; defined data-mart standards and reviewed designs, code, and test plans.

Applied Data Science Project | Saint Peter's University Sep 2020 - May 2021

- Built a customer-review sentiment classifier for a faculty-led Kmart project using Python, Pandas, and scikit-learn, identifying positive and negative feedback and operational themes such as delivery speed and logistics quality.

SKILLS AND CREDENTIALS

GenAI: PydanticAI, LiteLLM, Anthropic Claude, Amazon Bedrock AgentCore, LangGraph, prompt caching, structured outputs, DeepEval, OWASP LLM testing, LangSmith, OpenTelemetry
AWS: Lambda, API Gateway, DynamoDB, S3, SQS/DLQs, EventBridge, EKS, CloudWatch, Comprehend, IAM, VPC, KMS, Secrets Manager, Aurora PostgreSQL, OpenSearch
Engineering: Python, asyncio, Pydantic, pytest, Hypothesis, SQL, Terraform, Kubernetes, Jenkins, Git, CI/CD, Grafana, Dynatrace, Splunk, ServiceNow
Certifications: Claude Certified Developer, Foundations · AWS Certified Developer - Associate · AWS Certified Cloud Practitioner · Microsoft Certified Azure Fundamentals
Projects: CloudArchitect AI (LangGraph AWS architecture generator) · AlarmBrain (GenAI CloudWatch incident analysis) · uwaismansuri.com

EDUCATION

M.S. Data Science, Saint Peter’s University, 2021 · B.Tech. Computer Engineering, Indus Institute of Technology, 2019